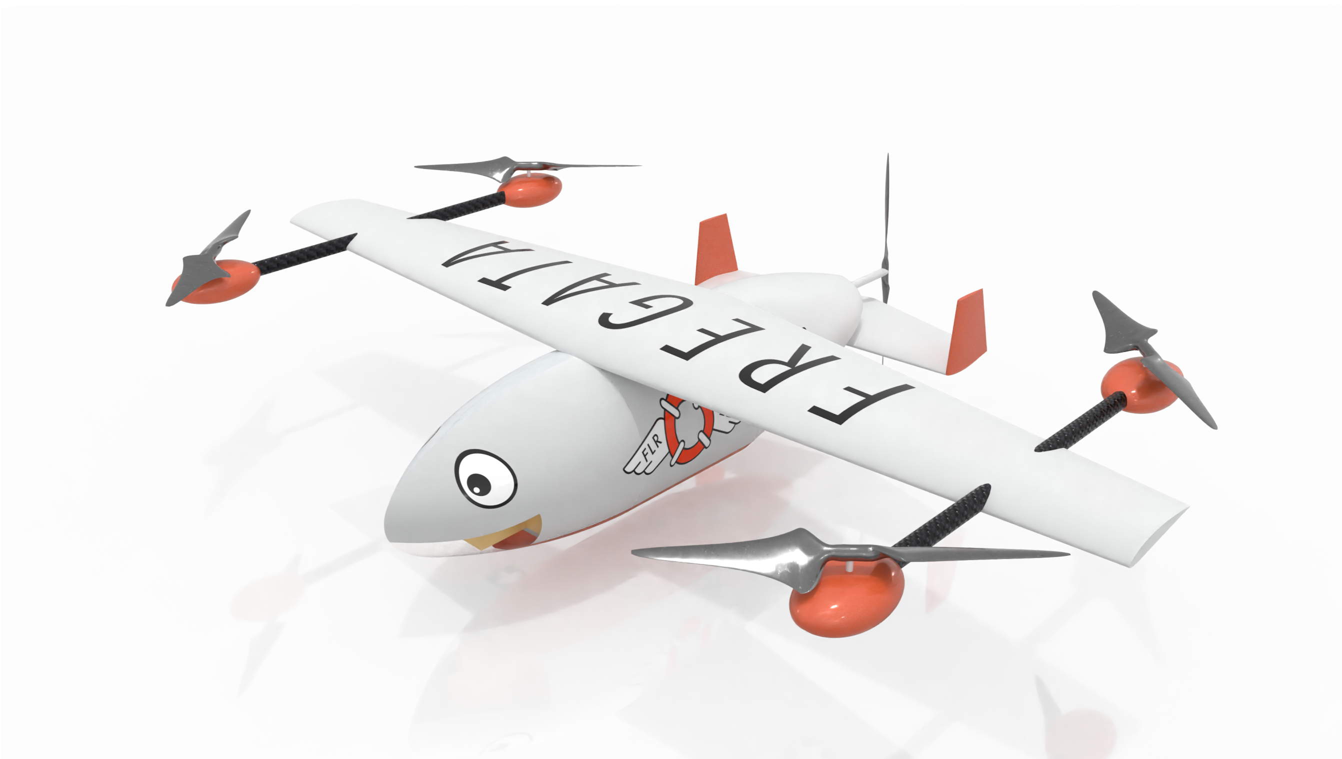
FLR22 - Flying Liferaft

Path planning with thermal detection for maritime search and rescue operations



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Abstract

Maritime accidents still lead to thousands of disappearances and deaths every year. Current coastal and NGO search and rescue services are over-strained and can't always provide a timely response. In this GDP we have developed Fregata a hybrid UAV capable of quickly providing first aid to people at sea. This fully autonomous system provides a first line of defence to drowning and hypothermia, increasing the survivability of the rescuees until surface vessels arrive at the scene. This report jointly focuses on the development of thermal-based detection capabilities, and on the determination of an optimal search path. Heuristic paths are considered through Monte Carlo simulations, and two novel paths are introduced. Optimal search heights, flight speeds and path parameters are studied and gradient descent optimisation is implemented. The optimal search height is found to be at 341 m with an optimal flight speed of 42.2 m/s.

Keywords - Path Planning, Optimisation, Thermography, Aerial Detection, Search and Rescue

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List of Nomenclatures

Symbol	Description	Units
α	Learning rate	-
β	Momentum parameter	-
λ	Wavelength	μm
Δt	Time step	s
ε	Emissivity	-
θ	Camera tilt angle	rad
ω	UAV's turning rate; angular velocity	rad/s
ψ	UAV's azimuth angle	rad
$\mathcal{W}$	Workspace	
$\mathcal{O}$	Obstacle region	
γ	Discount factor	-
σ	Trajectory	
Γ	Path parameter multiplicative factor	-
$C_{payload}$	Payload capacity	
CS	Converging spiral	
CSP	Commence search point	
CV	Computer vision	
d_c	Target characteristic dimension	m
ES	Expanding spiral	
f	Focal length	mm
FOV	Field of view	deg
FS	Field size	m
FPS	Frames per seconds	s^-1
GMM	Gaussian mixture model	
GSD	Ground sample distance	m
h	UAV's height	m
IFOV	Instantaneous field of view	mrad
IR	Infrared	
J	Objective function	-
LKP	Last known point	
LWIR	Long-wave infrared	
MSAR	Maritime search and rescue	
ML	Machine learning	
N	Number of targets	-
N_{saved}	Number of targets saved	-
n_{max}	Maximum load factor	-
$q = [x, y, h, \psi]$	UAV's configuration	[m, m, m, rad]
PIW	Person-in-water	
POC	Probability of containment	-
POD	Probability of detection	-
POS	Probability of success	-
PS	Parallel Sweep	
SAR	Search and rescue	
SNR	Signal-to-noise ratio	-
SRU	Search and rescue unit	
SSR	Spot size ratio	-
TIR	Thermal infrared	
TTT	Time-to-target	
UAV	Unmanned aerial vehicle	
UAS	Unmanned aerial system	
v	UAV's linear velocity; flight speed	m/s
VS	Sector search	

Subscript

$(\cdot)_H$	Relative to horizontal/lateral dimension
$(\cdot)_V$	Relative to vertical/longitudinal dimension
$(\cdot)_\ell$	Relative to loiter phase
$(\cdot)_k$	Relative to iteration k

1 Introduction

As widely reported, migrant crossings of the Mediterranean sea lead to great loss of life, with at least 24'176 people disappearing in Mediterranean waters since 2014 [1]. Search and rescue (SAR) resources of the neighbouring countries are thus understandably strained to cover such an extensive area, while aiming to provide a fast response time. Coast guard boats prove to be slower than desired, and helicopters require expensive maintenance and expert operation while only being capable of rescuing a limited number of people. This GDP developed a fully autonomous hybrid (rotary-wing and fixed-wing) unmanned aerial vehicle (UAV) capable of providing timely aid to a person at sea. The UAV was designed to achieve a time-to-target (TTT) of less than 30 min, over a 70 km range (selected during initial sizing and corroborated by Julian and Foster's reports). According to the defined mission profile, cruise can take up to 20 min, leaving 10 min for reconnaissance and payload deployment. Relevant specifications will be presented as needed, but for a broader understanding of the design please refer to the executive summary.

SAR operations aim to locate and provide aid to a person in distress as fast as possible. The faster search and rescue units (SRUs) reach the target, the more likely it is to survive. When it comes to maritime operations, TTT is especially crucial as drowning can be fatal in a few minutes, and hypothermia can lead to loss of life in as fast as 30 min [2]. Unlike in some ground/terrestrial SAR scenarios, in maritime SAR (MSAR) the target can drift, thus rendering the last known point (LKP) progressively less useful as time advances. Despite this, TTT is not the only factor, and SAR resources should not be allocated to a search without considering the time and monetary costs, and the risk rescuers are being placed in. UAVs are becoming increasingly more affordable and are gaining traction in many civilian areas. Apart from having a shorter setup time, UAVs present a cheaper SAR solution, which is not subject to fatigue and one where SAR operators are not placed into dangerous situations. They can also provide an aerial view of the accident scene to the rescue team.

The rest of this report is organised in the following manner: Section 2 provides a formal definition of the problem to be solved, Section 3 considers the efficacy of thermal-based detection, Section 4 discusses the determination of the search area, compares different heuristic search patterns and provides insight into determining the optimal path for a given MSAR scenario, Section 5 discusses limitations of the work, further steps and ethical problems raised and Section 6 relates the work presented back to the aims and objectives.

2 Optimisation problem definition

The term "targets" will be used to refer to the people in distress that must be located and provided aid to. The UAV has a modular payload that can either deliver 24 individual buoys with chemical-based heating to fight hypothermia and drowning, or a single liferaft capable of sustaining 6 people. Let N denote the total number of targets to be saved in a given MSAR scenario. This number, or at least an estimate of it, can be provided by those who located the accident. Similarly, as will be further discussed in Section 4.1, an approximate location of the accident is also assumed to be provided and the targets are taken to be uniformly distributed over the square search area. A random variable $N_{saved}(t)$ is defined to represent the number of targets saved up to a given instant t. This is a random variable since the number of people saved depends on their initial distribution along the search area, the commencement search point (CSP), their physical condition, the weather conditions, the amount of SRUs deployed, the time elapsed between the start of the accident and the detection of said predicament by onlookers, the distance between the accident and the current location of the SRUs (dictates cruise duration) and the quality of the information provided to the SRUs. For the purposes of this report, we are only concerned with the behaviour of a single aircraft, neglecting resource allocation and other SAR considerations. t_0 denotes the start of the search (when the vehicle enters the search area) and t_f the end of the search (once all targets have been saved or when fuel/energy have been fully depleted), such that $t_f - t_0 = \Delta t_{search}$ is the total search duration.

The vehicle workspace is $\mathcal{W} = \mathbb{R}^3$, as the problem's geometry is clearly 3-dimensional. The obstacle region is denoted by $\mathcal{O}(t) \subset \mathcal{W}$ and it defines the locations of the workspace occupied by obstacles (i.e. no-fly zones, areas occupied by other aircraft or boats, ...). $\mathcal{O}$ is a function of time, as the obstacles need not be stationary. A candidate trajectory is denoted by $\sigma \in \Sigma$, where Σ corresponds to the set of all possible trajectories. $\sigma(t)$ describes the vehicle's configuration q at time t, where $q = [x, y, h, \psi]$. A trajectory is effectively a path parameterised through time. The goal of the optimisation problem is to find the optimal trajectory $\sigma \in \Sigma$ which leads to the minimisation of the objective function J:

$$\min_{\sigma \in \Sigma} \quad J(\sigma) = \frac{1}{N} \cdot \int_{t_0}^{t_f} (N - \mathbb{E}[N_{saved}(t)]) \cdot \gamma(t, \kappa)\ dt \tag{1}$$

$$\text{where} \quad \gamma(t, \kappa) = 1 - \exp\left(\frac{-t}{\kappa}\right) \tag{2}$$

$$\text{s.t.} \quad \int_{t_0}^{t_f} BSFC(t) \cdot P(t)\ dt \leq M_{fuel,\ell} \tag{3}$$

$$\int_{t_0}^{t_f} E(t)\ dt \leq E_{battery,\ell} \tag{4}$$

$$\sigma(t) \in \mathcal{W}_{free} = \mathcal{W} \setminus \mathcal{O}, \quad \forall t \tag{5}$$

$$F(q, t) = 0 \quad \text{and} \quad F(q, \dot{q}, t) = 0, \quad \forall t \tag{6}$$

$$C_{payload}(t_f) \geq 0 \tag{7}$$

Defining J is the most crucial step of the optimisation process. Care should be taken to consider the different ways in which a given objective function might be minimised, to ensure that these all lead to a better solution of the problem at hand. An ill-defined objective function will lead to solving a different problem than that intended. Before justifying the chosen definition of J shown in (1), let us consider some possible alternatives. A naive goal for J could be to maximise the spatial-temporal coverage of the search area. The clear solution would be a path at a high altitude to maximise the ground area covered by the camera, and at high speed to maximise the area covered per unit of time. Regardless, this does not take into account how the probability of detection changes as we alter the flight altitude and/or velocity. One could also consider maximising the number of targets saved, however SAR operations are time-critical and as such a timely path should be incentivised. Note that the value of J is a function of the path σ taken. $\mathbb{E}[N_{saved}(t)]$ is used to account for N_{saved} being a random variable. The objective function integrates the difference between the total number of targets in the scenario and the number of targets that have already been saved at a given instant, over the duration of the mission. That is, the number of people still left to save $N - N_{saved}(t)$, at a given instant t, is integrated over time. A discount factor γ is applied to the integrand and the result is then divided by the number of targets, to normalise the objective function (which would otherwise also clearly be a function of N). Observe that for a given number of targets N (not a function of time) and for a constant discount factor γ, the objective function can be expanded into $\frac{\gamma}{N} \cdot (\int_{t_0}^{t_f} N\ dt - \int_{t_0}^{t_f} \mathbb{E}[N_{saved}(t)]\ dt) = \gamma \Delta t_{search} - \frac{\gamma}{N} \int_{t_0}^{t_f} \mathbb{E}[N_{saved}(t)]\ dt$. The objective function can thus be minimised by making the search mission shorter or by increasing $\mathbb{E}[N_{saved}(t)]$. From the definition of t_f, the former implies saving all the targets in a shorter time-frame (fuel/energy are already sized and shouldn't change) and the latter implies saving more people in the same duration, the same number of people in a shorter duration or a combination of both.

$\gamma(t, \kappa)$, defined in (2), is a discount factor designed to incentive different behaviours depending on the type of payload the aircraft is carrying. When carrying buoys we have $\kappa = 0$, which leads to $\gamma(t, \kappa) = 1$ and as such this term has no effect on the objective function (multiplicative identity). When carrying a liferaft $\kappa \in \mathbb{R}_{>0}$, and so the discount factor lies within the interval $\gamma \in (0, 1) = \{\gamma \in \mathbb{R} \mid 0 < \gamma < 1\}$. No theoretically optimal values of κ are prescribed but instead one could take it to be an experimentally determined parameter that would intuitively change value depending on the specific MSAR scenario at hand. κ would intuitively take larger values for calmer weather conditions and more favourable target conditions (they already possess life-jackets or the water temperature is warmer). The greater κ is, the less penalising the initial seconds of the search become (as κ increases a longer duration is discounted for a

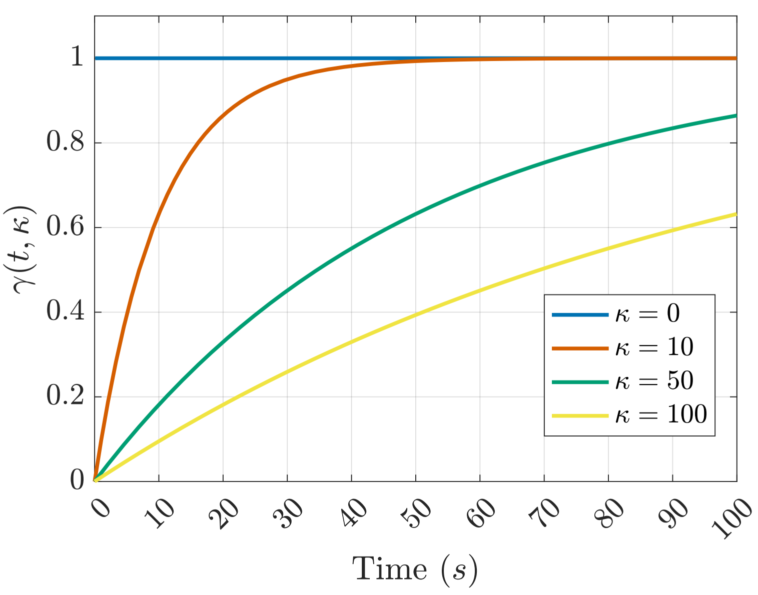

Figure 1: Discount factor γ along time for different values of κ

larger amount). For the individual buoy case, there is a clear incentive to drop a payload as soon as a target is detected. Ignoring a target to search for a larger group would provide no upside when it comes to minimising J or saving more people. However, for the liferaft case, since there is a single deployment and the payload can save multiple people at once, it might be beneficial to ignore a target and keep searching in the hopes of finding a larger number of targets close together. γ serves to incentivise exploration closer to the start of the mission. Figure 1 shows the effect of κ on the discount factor across time.

(3) implies that throughout the duration of the search phase, the vehicle must not consume more fuel mass than that which was allocated for loiter. The brake specific fuel consumption $BSFC(t)$ along with the power produced at a given instant $P(t)$ are used to determine the instantaneous fuel consumption of the vehicle. Thanks to data provided by Farsai and Jordi, along with collaboration from the Aero team, fuel consumption rates for different flight velocities were calculated. $M_{fuel,\ell}$ denotes the total fuel mass allocated for the loiter phase. (4) similarly constraints the battery energy usage where $E(t)$ is the instantaneous energy consumption by the VTOL propellers and $E_{battery,\ell}$ the battery capacity allocated for loiter (with values provided by Natalia). While energy is usually included in the objective function to maximise efficiency, for our specific scenario it is inconsequential how much energy/fuel we come back to base with, as timeliness is paramount. (5) indicates that trajectories should only transverse the free space (not collide with any obstacle).

(6) represents the dynamic constraints of the UAV motion (set by maximum turning rate, maximum climb rate and maximum descent rate). Values limiting the aircraft's motion were provided by the Structures team and Kristoff. Like a car, a fixed-wing UAV is a fundamentally non-holonomic system [3, 4] (an omnidirectional robot is an example of an holonomic system). Despite it being capable of occupying any pose (position and orientation) in the workspace $\mathcal{W}$, an UAV's current configuration is a function of past ones and only some types of motions are allowed from a given current configuration. Non-holonomic constraints, also called velocity constraints, thus do not affect the reachable configurations of the system but solely which configurations can be directly reached from other ones. Constraint (7) ensures that the payload capacity is not exceeded (we don't drop more buoys or liferafts than those we are carrying).

It is not tractable to explore all σ which satisfy the constraints above. Optimal path planning is an NP-hard problem [5], therefore approximate/heuristic methods are relied upon to find a solution. The environment tackled in this report has unknown goal positions, further complicating optimisation. While methods mentioned in Section 5.2 might explore a wider variety of trajectories, heuristic paths were considered.

The equations above were based on the continuous temporal and spatial domains in which the UAV will operate. It must be noted that the simulations discussed in Section 4 were run on the discrete equivalents of these equations. This is the case as computers are fundamentally discrete machines incapable of performing operations on continuous values. The discretisation of the domains should not have introduced significant errors, as small step-sizes were chosen and finite differences were used as appropriate.

3 Aerial thermal imaging

3.1 CV and Thermography

Computer Vision (CV) is a major branch of computer science that develops techniques to enable machines to gain understanding of a scene from digital images. There are significant disparities between the CV problems that state-of-the-art methods are tested and evaluated on, and those of a thermal infrared (TIR) MSAR operation (thus transfer learning is nonviable). Standard CV methods deal with images taken relatively close to the target (it occupies a significantly greater number of pixels), through a visible light camera (which provides colour and texture information lacking in TIR, and has higher spatial resolution) and from a ground-based viewpoint (which changes the object shape). Images obtained from aerial TIR will present humans in the form of small blobs, due to the larger distance between camera and target, the oblique viewpoint and because only the non-submerged part of the body will be visible. This change of human shape and lack of discernible features greatly complicates the application of conventional methods, and makes distinction between humans and other hot objects quite complex (fortunately the maritime scenario will rarely contain other hot objects). Due to the speed at which the drone will be travelling, and it being subjected to gusts and other disturbances, motion blur will also become an issue.

A long-wave infrared (LWIR) thermal camera was chosen for this application since humans emit thermal radiation more strongly in the 8-14 μm range [6, 7]. Visual/Conventional cameras have excellent spatial and temporal resolutions at an affordable price. However, they are not suited to operate under all weather conditions. Conventional cameras are useless in no-light scenarios (as they rely on the visible spectrum light reflected by surfaces to form images) and have subpar performance in low-light scenarios. Infrared (IR) cameras are more expensive due to economies of scale, an almost exclusive use in military applications until recently and the need for more exotic materials (camera lenses must be made of germanium, as the glass used for visual cameras is IR opaque). The lower resolutions are a byproduct of both economical and physical reasons, as the longer wavelengths of IR require larger sensors to absorb enough radiation for measurements. Thermal images have a lower signal-to-noise (SNR) ratio than visible images. Fortunately, aerial maritime images present far less background clutter than most conventional ground images. Like other homeothermic endotherms [8], humans will appear as bright blobs providing clear thermal contrast in a maritime setting (ocean water has a uniform temperature [7], making objects of interest stand out more). Waves crests and sun shades affect TIR images considerably less. TIR readings are significantly less affected by fog and mist [6]. Still, water vapour, snow or rain cause problems due to their strong IR absorption.

Since the goal is target detection, absolute temperature measurements are not required as relative measurements are enough for foreground segmentation. IR physics can provide valuable insights into the limitations of thermography. The camera must be configured with an appropriate emissivity value, $\varepsilon = 0.98$ for humans, in order to obtain reasonable readings. Unfortunately, emissivity varies with viewing angle θ, and for $\theta > 45°$ it stops being constant and starts varying non-linearly [9]. Despite enabling the coverage of a wider area (both longitudinally and laterally), higher viewing angles imply that the distance between the target and the camera will be larger (for a constant altitude) and as such lead to lower spatial resolutions and higher levels of atmospheric IR absorption (although this report does not consider how atmospheric absorption impacts detection rates at different altitudes). As such, the camera was taken to be pointing vertically downwards $\theta = 0°$ (no tilt-angle). Pointing directly downwards also ensures that the horizon and sky don't appear in the frame, thereby simplifying the detection procedure. Additionally, if the camera were to be tilted, then every pixel would correspond to a differently sized area on the ground (thereby complicating resolution calculations). Thermography uses the infrared radiation emitted by an object's surface to determine its temperature, but surface temperature differs from internal temperature and, despite being less prevalent than in the visible spectrum, reflection can still occur in IR. Additionally, clothing reduces the heat emitted making humans blend in more with the environment. Additional comments on camera resolution calculations can be found in Appendix A.

The GC3-T camera model by Swell-Pro was selected. It retails around 6000£, with specs listed on Table 1. It has a decent waterproof rating, which was quite rare for drone thermal cameras within this price range. Besides, it comes with a low-light FHD visible camera and a stabilising gimbal. The visible spectrum camera will not be considered but it presents great possibilities for further development. The sensitivity value mentioned corresponds to relative temperature measurements (absolute temperature sensitivity is orders of magnitude worse). The purchase of a separate lens was also considered but neglected due to their expensive nature (upwards of 2000£), the increased dimensional footprint the camera would take on the fuselage and an inability to find a zoom lens suited for TIR cameras (despite digital zoom bringing no value when it comes to detection, a lens with optical zoom would allow a wider field of view (FOV) for high area coverage and a narrower FOV upon request for clearer identification). GC3-T comes with $f = 19$ mm lens, which has a FOV of 22x18°. All cameras within the considered price range are equipped with a thermal detector (instead of a photon one) and thus don't require detector cooling (which leads to the lower cost). As pointed out in [10], the nature of an uncooled detector leads to camera prices increasing significantly with f size, therefore reinforcing the unfeasibility of acquiring a telephoto lens.

Table 1: GC3-T Specifications.

Parameter	Value
Detector Type	VOx microbolometer
Sensor Resolution (px)	640 x 512
IFOV (mrad)	0.60 x 0.61
Spectral Band (μm)	8-14
Frame Rate (Hz)	50
Waterproof rating	IP67
Sensitivity (mK)	≤ 50

Having defined the camera specifications, let us consider the effect of search altitude h on the detection rates. An initial analysis based on TIR theory reveals that a target should take up at least 3x3 pixels on the image plane for an accurate thermal measurement (if the target is less than 3x3 pixels large, then the measured temperature will be a blend with the temperatures of surrounding objects) [11]. Notwithstanding, the difficulty lies not in obtaining an accurate thermal measurement, but in having enough detail to classify the object correctly. From literature, 5x5 pixels are sufficient if humans are the only expected target or false positives are not a major concern [12, 13].

3.2 Johnson's Criteria

The various CV tasks have different levels of difficulty and algorithm maturity. Detection relates to the ability of distinguishing an object of interest from the background of an image (detect that something is present). Recognition relates to being able to classify the object in the image into a set of pre-determined classes (human, boat, ...). Identification involves being able to differentiate between objects of the same class. DORI [14] (Detection, Observation, Recognition, Identification) is a widely used standard to determine spatial resolutions required to perform these tasks. It is however specific to visual images. Thankfully, Johnson's Criteria [12, 15, 16] which inspired DORI, was defined for TIR images. Johnson introduces a factor N_{50} which describes the number of cycles required for a 50% level of performance (i.e. 50% detection probability in our case). For detection we get $N_{50} = 0.75$. Considering the average human shoulder width, the likely vertical position of a person in water (PIW) and the oblique viewpoint of an aerial thermal image, a target size of 0.5x0.5 m was chosen, which is corroborated by literature [12, 13]. Using this, we can determine the target's characteristic dimension $d_c = \sqrt{\text{Target}_H^2 + \text{Target}_V^2} = \sqrt{0.5^2 + 0.5^2} = 0.5$ m. Then, the number of cycles across the target N is given by $N = d_c/(2 \cdot GSD_{avg})$ where GSD_{avg} is the average ground sample distance (GSD). Please refer to Appendix A for a review of camera physics. Succinctly, GSD corresponds to the area covered by a single pixel. The higher GSD is, the lower the spatial resolution and the fewer cycles over the target we will have. Lens distortion effects have been ignored in this analysis. Johnson's Criteria allows us to obtain an estimate of the probability of detection $P(N)$ given a certain number of cycles N

$$P(N) = \frac{(N/N_{50})^{2.7+0.7\cdot(N/N_{50})}}{1 + (N/N_{50})^{0.27+0.7\cdot(N/N_{50})}} \tag{8}$$

Using (8) and recalling the relationship between N, GSD and flying altitude h, explored in Appendix A Equation (28), we can obtain the probability of detection of the human target as a function of altitude (shown in Figure 2 alongside the probability of recognition and identification).

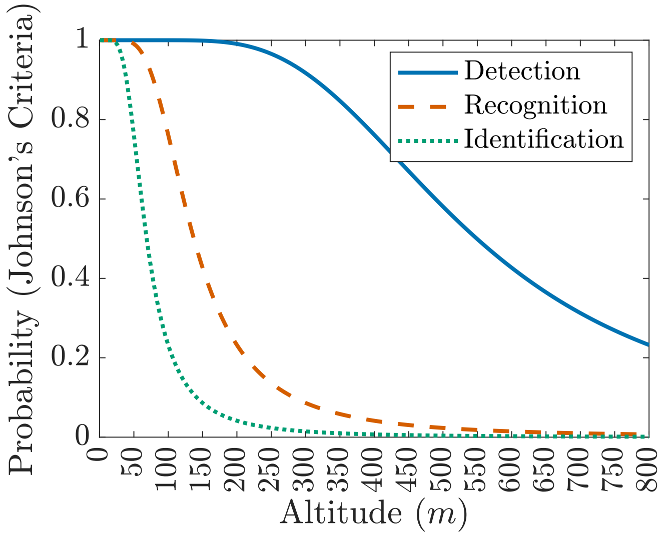

Figure 2: Probability of detection, recognition and identification with altitude from Johnson's Criteria, for camera with GC3-T specs

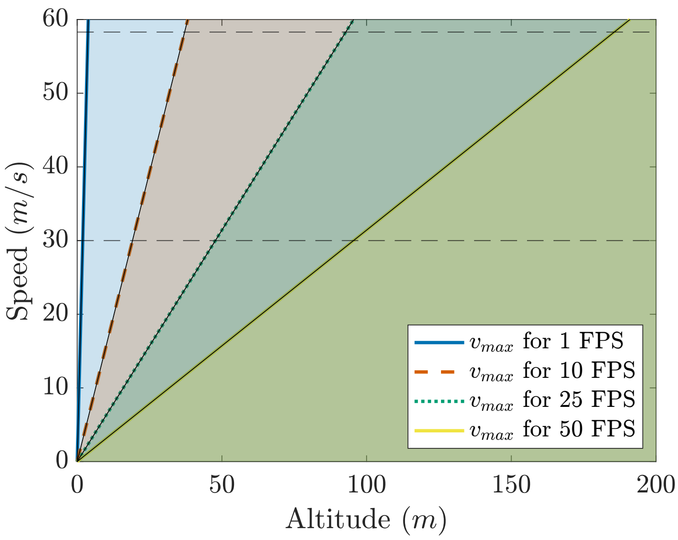

Figure 3: v_{max} with altitude for full coverage of sea surface at different FPS

While d_c influences the choice of flight altitude, the camera's frame-rate affects the choice of flight speed. We don't want to travel too fast that we miss parts of the ground surface between consecutive frames or too slow since that is detrimental to the MSAR goal. As described in Appendix A, field size defines the total ground area covered by the camera. $\text{FS}_V(h)$ defines the longitudinal dimension of the surface covered at a given altitude (and $\text{FS}_H(h)$ the lateral one). For full coverage of the water surface, the flight speed must be restricted such that $v \leq \text{FS}_V/(1/FPS)$ where FPS is the camera's frame rate. If we wish to obtain more frames of a given area (for redundancy and object tracking), then the maximum speed becomes $v \leq \text{FS}_V/(frames/FPS)$, where $frames$ indicates the number of frames desired per second. From Figure 3 it is clear that the camera's frame-rate will not act as a bottleneck, and detection will instead be constrained by the algorithm's processing time.

3.3 Detection algorithm and Target Backprojection

Despite no particular detection algorithm being evaluated in this report, due to time constraints, some possible solutions are discussed in this section. Fully automated detection systems are considered, as these are more reliable and accurate than those requiring operator interaction. Broadly, candidate algorithms can be broken down into traditional and machine learning (ML) based approaches [17, 18, 19, 20], and into two-step or one-step approaches. Due to the limited computational resources available onboard the UAV, care should be taken to analyse not only an algorithm's accuracy but also its inference time (as real-time human detection is the primary goal). Here the low resolution of TIR images is beneficial, as besides being single channel (while RGB are 3 channel) the smaller number of pixels leads to faster computations.

A one-step detection approach directly obtains the objects' positions in the image plane from the captured image. These are often complex and less accurate due to their end-to-end nature. A two-step approach involves first separating the foreground from the background and then classifying the foreground objects to determine which ones correspond to targets. The first step can be performed through the generation of saliency maps [7, 21, 22], by background subtraction (background might be dynamically generated from in-flight data or given apriori) [23, 24], using Gaussian mixture models (adaptable to different backgrounds and can capture the bimodal nature of the thermal radiation in MSAR scenarios - radiance reflected from sky and emitted from sea) [25, 26] or via gradient-based blob extractions [27, 28]. The second step can be performed via SVMs [28], HOG detectors [24] or other any other classification method. Despite being more computationally intensive, a two-step approach is recommended for this project due to its modularity and simplicity.

Traditional computer vision approaches rely on decades of heuristics and mostly edge-based techniques. In contrast, ML which clearly dominates the CV landscape nowadays, relies almost exclusively on obtaining large amounts of data on which to train on. The performance of a given ML application is intrinsically related to the quality of the training data (how closely it matches the distribution of the real data, label quality, extensiveness of edge cases, ...). Unfortunately, there aren't many datasets of aerial TIR images obtained by drones from which one could train an algorithm on, but it would be possible to start operating with a traditional detection algorithm and slowly train an ML algorithm with the data collected from missions. When training an algorithm for MSAR operations, the most relevant metric to monitor is by far the algorithm's Recall, where Recall = TruePositives / (TruePositives + FalseNegatives). While false positives will only lead to a small loss in time and energy, a false negative might imply a loss in life. Traditional algorithms should be initially adopted, and data captured during missions should be stored for future developments.

Upon successful detection of the object in the image, we can easily obtain its coordinates in the image plane (pixel coordinates). However, for a successful deployment, which was defined by Foster to be one with the payload dropping within 5 m of the target, we actually require an estimate of the target's pose relative to the UAV's pose (world coordinates). A given pixel in the image plane can be related, through basic trigonometry, to a ray of points in the world (given we know the internal camera matrix, which can be obtained via calibration). If we assume that the targets are located at the plane $h = 0$ (implies that the water surface is flat and located at 0 altitude, reasonable assumption for calm weather conditions), we can obtain an estimate of the world coordinates of the target by calculating the intersection between the ray projected by the pixel of interest and the plane $h = 0$. This can then be related to the UAV's frame of reference by knowing the camera's position in the UAV.

4 Path Planning

4.1 Search area delimitation

A critical design driver is the size of the area to be searched by the UAV. Using results from [29], where distance measurements taken at sea using binoculars were found to have a 9% error at around 9 km, the initial uncertainty in the location of the targets was taken to be 800 m (assuming that those that detect the accident have their own GPS coordinate, they can measure the distance between themselves and the migrants to within 800 m).

Using this initial localisation error and average values for wind and current speeds (wave effects were neglected), a particle-based drift model [30, 31, 32, 33] was applied to determine how much the targets would spread by, over the 20 min that cruise takes. The highest wind speed values in the Mediterranean are of 11-12 m/s [34] and the highest average currents speeds occur during the winter at 0.84 m/s [35]. An OpenDrift model [36] was ran at 37.4° latitude and 18° longitude using the values mentioned above (with 3 m/s wind uncertainty and 0.2 m/s current uncertainty) with 10'000 particles. The uncertainty values alongside the large number of particles should provide a fairly conservative estimate for the area size. It is not unrealistic to assume that the system will have access to current data (which is predictable) or wind data (in the form of forecast or through real-time measurements). The object type was chosen to be a conscious person-in-water (PIW) in a vertical position (note that different objects have different drift coefficients). An illustration of the OpenDrift simulator is shown in Figure 4. Through analysing the spread of the particles after 20 min, a search area of 1200x1200 m was determined. For simplicity of path planning, it was set to have a square shape. This analysis also made clear that any drift that might occur during the search phase would be negligible (due to its shorter duration) and so targets were modelled as stationary during the search.

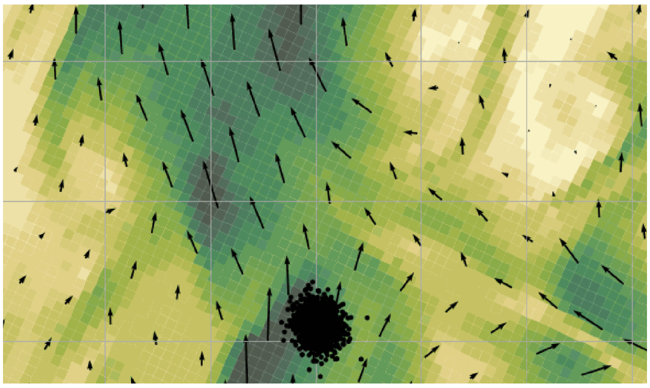

Figure 4: Particle based drift modelling using winds and currents on OpenDrift

4.2 Probability of Success

In search theory literature, multiple metrics have been developed to aid in the evaluation of different search paths. We will consider the probability of containment (POC), probability of detection (POD) and probability of success (POS) due to their overwhelming popularity [31, 32, 33, 37, 38, 39], and use them to determine a preliminary optimum flight altitude.

POC is the probability that the target is present in a given area. As done in literature, it is assumed that the totality of the search area has POC = 1 (i.e. all targets are within the search area). For a given flight altitude the corresponding FS can be obtained using the methods explained in Section 3. Assuming that the targets are uniformly distributed over the search area, then the POC of the area seen by the camera at a given instant is given by POC = $\text{FS}_H \cdot \text{FS}_V$ / total search area. Targets were taken to be uniformly distributed over the search area. This probability distribution is best suited for when little to no data is known about the target [37, 40], and so it acts as a conservative estimate. A normal distribution around the LKP could also be considered, and it would greatly simplify search planning, but this was opted against as these typically require two or more independent position estimates. A line distribution was neglected as the intended target path is neither known nor likely to be followed (as the target's movements are assumed to be heavily restricted by the predicament it is found in).

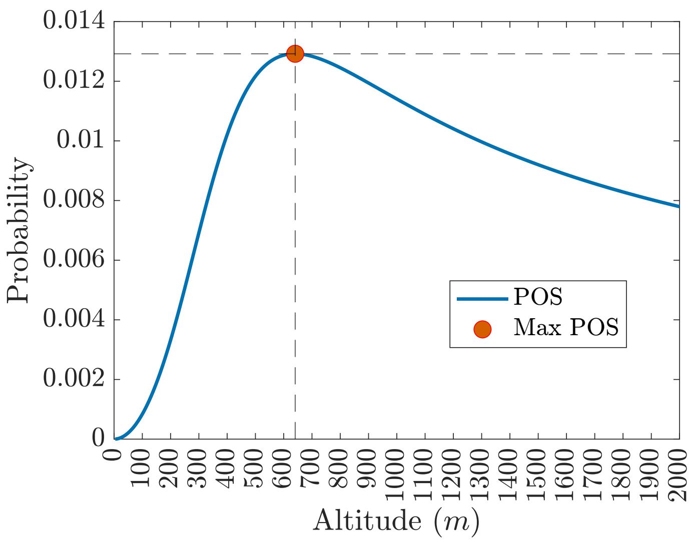

Figure 5: POS with altitude

POD determines how likely a target is to be found if it is within the range of the sensors. While usually defined via a normal distribution as a function of the lateral distance between the SRU and the target, in our particular case the POD can be obtained as a function of flying altitude via Johnson's Criteria. When a given target is located within the FOV of the camera, the probability of its detection is taken to be solely a function of its apparent size and not its position on the image frame.

Finally, POS defines the probability of finding the search object. It is the ultimate measure of search effectiveness. It can be calculated via POS = POC · POD, and its maximum value is shown in Figure 5 (note that the value shown is the POS for a given FS(h) area, not of the whole mission). Figure B.1 shows POC and POD as a function of flying altitude. With increasing altitudes, the FS increases and consequently so does the POC, but the POD decreases following Johnson's Criteria. The height at which the probability of detection is maximised was found to be 640 m. This preliminary value will kick-start the analysis.

4.3 Heuristic Paths

Much of the literature on UAV path coverage focuses on swarm coordination and optimal allocation of multiple resources [41]. This report is solely concerned with single-agent behaviour, which is the overwhelmingly likely scenario according to our business model. For the purposes of this section, obstacles were ignored.

The UAV was modelled as a non-holonomic point mass moving according to the following equations

$$\psi_{k+1} = \psi_k + \omega_k \cdot dt \tag{9}$$
$$x_{k+1} = x_k + v_k \cdot \cos(\psi_{k+1}) \cdot dt \tag{10}$$
$$y_{k+1} = y_k + v_k \cdot \sin(\psi_{k+1}) \cdot dt \tag{11}$$
$$h_{k+1} = h_k \tag{12}$$
$$\text{s.t.} \quad |\omega_k| < \omega_{max} \tag{13}$$
$$h_{min,safety} \leq h_k \leq h_{ceiling} \tag{14}$$

where it follows that $\dot{\psi}_k = \omega_k$ and $[\dot{x}_k, \dot{y}_k] = v_k$. Energy and fuel capacity were updated at every time-step and the payload capacity constraint was also monitored. (13) encompasses the dynamic constraints, as altitude is kept constant throughout the path.

Through decades of research and field experiments, a few heuristic search patterns have emerged as dominant in MSAR operations [31, 33, 37, 42]. Here we will focus on Parallel Sweep (PS) and Sector Search (VS), which were slightly modified to respect the dynamical constraints of the non-holonomic system and present two novel search patterns, the Expanding Spiral (ES) and the Converging Spiral (CS). Other prevalent search patterns were discarded for the following brief reasons: expanding square search does not respect the dynamical constraints requiring sharp 90° turns; track line search is used when a target disappears while en route to a known destination; contour search is effective when dealing with elevation changes in the terrain. The search patterns will be briefly described in words and through diagrams, but for a better understanding the author recommends viewing the short clip in Figure 6 (https://vimeo.com/721814024). All the paths are parameterised by a single variable. There's often an intuitively optimal value for said parameter and $\Gamma_{path\ name}$ will be used to indicate the ratio between the parameter value being tested and the intuitively optimal one, such that $\Gamma_{path\ name}$ = current parameter value / intuitive value.

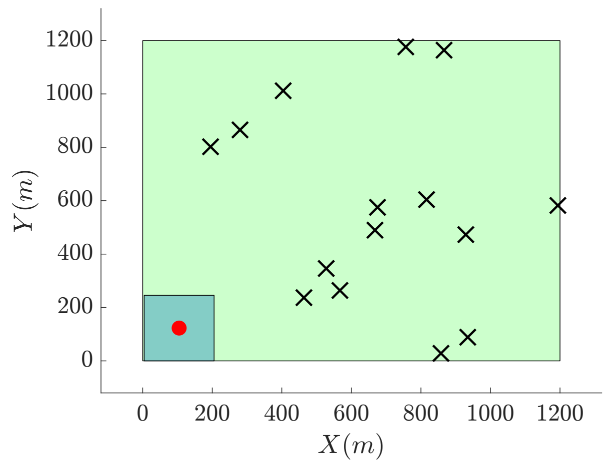

Figure 6: Heuristic Paths Video: https://vimeo.com/721814024

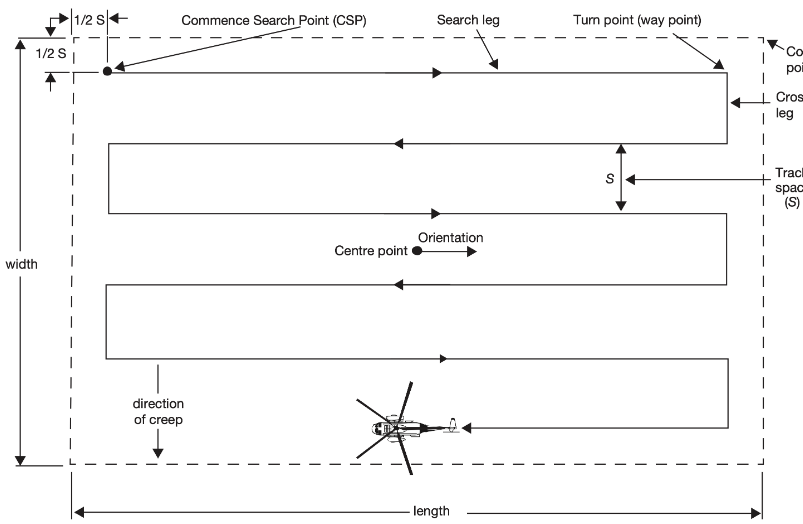

Figure 7: Unmodified PS path. Image from [37]

PS, shown in Figure 7, is deployed whenever there is a large uncertainty in the targets' whereabouts. It covers the area uniformly, thus being particularly effective over flat terrain. Since the search area in question is square and not rectangular, PS is also effectively a Creeping Line path (where the movement is done along the shorter side of the rectangle). The traditional PS was modified to respect the dynamical constraints of the UAV (turns were changed from 90° to those permitted by the maximum turning rate). The turning radius is denoted by R and the track space (distance between parallel search legs) is denoted by S. Figure 8 shows the relation between the two. For full coverage of the search area, it must hold that $S \leq \mathrm{FS}_H$ (recall FS_H was the horizontal ground distance covered by the camera). Since from the dynamical constraints of the turn we know that $S/2 = R$, where R is the turning radius, it can be said that for full coverage $\mathrm{FS}_H \geq 2R = 2v/\omega$. Recalling that FS_H is solely a function of flying altitude and that the maximum turning rate is given by $\omega_{max} = (g\sqrt{n_{max}^2 - 1})/v$, then one can state that for a given flight altitude h, the maximum linear velocity v that enables a full coverage PS path is given by

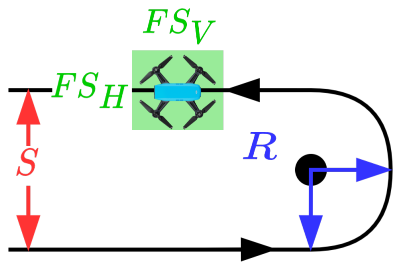

Figure 8: Relationship between S and R on modified PS path

$$v_{max} = \sqrt{\frac{\left(\mathrm{FS}_H(h) \cdot g\sqrt{n_{max}^2 - 1}\right)}{2}} \tag{15}$$

This relationship is shown in Figure 9. Intuitively a value of $S = \mathrm{FS}_H$ would provide full coverage without any overlap in the observed area. As such this is taken as the baseline value to be analysed in the following section (and Γ_{PS} indicates a ratio relative to it).

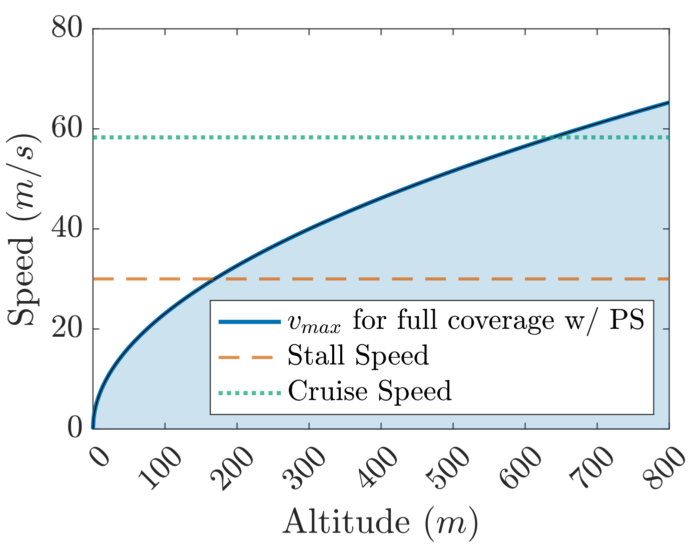

Figure 9: v_{max} with altitude for full coverage on PS path

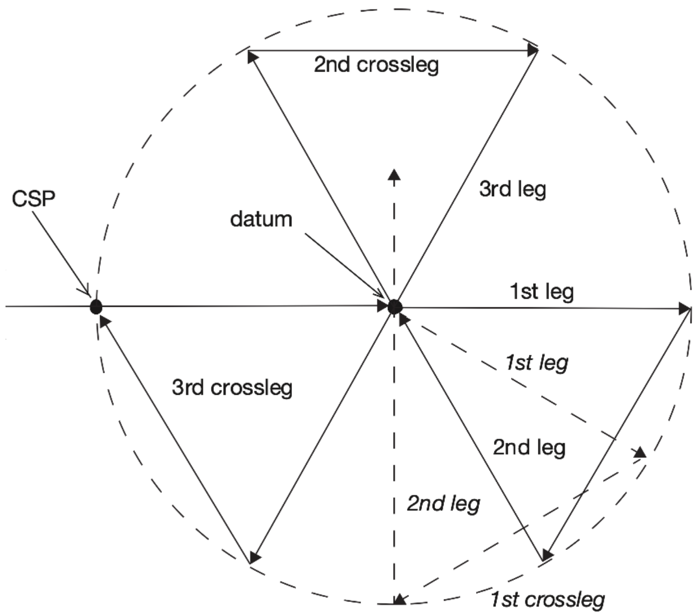

Figure 10: Unmodified VS path. Image from [37]

VS, shown in Figure 10, is effective when the target LKP is known and the search area is relatively small. Since VS relies on a circular search area, for this path alone, the area was considered to be a circular one which is inscribed by the 1200x1200 m square area. As seen in the video, VS covers the central area particularly well, making it well suited to deal with a point distribution (normal about LKP). In literature, all turns are commonly made starboard with a 120° turning angle but due to minimum radius constraints this proved to be too small. 135° was taken as the intuitive value.

While expanding square search patterns exist in literature, these don't satisfy the non-holonomic constraints of the UAV. Thus novel spiral patterns were developed. An Archimedean spiral (or arithmetic spiral) was picked to define the motion due to its nice property of having rays originating from the origin intersecting successive turnings with a constant separation distance. In polar coordinates (r, θ) an Archimedean spiral is described by

$$r = a + b \cdot \theta \tag{16}$$

where a dictates the centerpoint's distance from the origin and b the distance between consecutive turnings (if θ is in radians, successive turns are $2\pi b$ units apart). Figure 11 illustrates an Archimedean spiral. The ES pattern first moves towards the centre of the search area and then executes an expanding Archimedean spiral defined by $a = 0$. An intuitive choice of b would be $b = \mathrm{FS}_H/2\pi$ such that full coverage is achieved without any overlap. In the CS pattern, the vehicle has a CSP at a corner of the square area and then enters an Archimedean spiral towards its centre where $a = 600$ and b has an intuitive value of $b = -\mathrm{FS}_H/2\pi$.

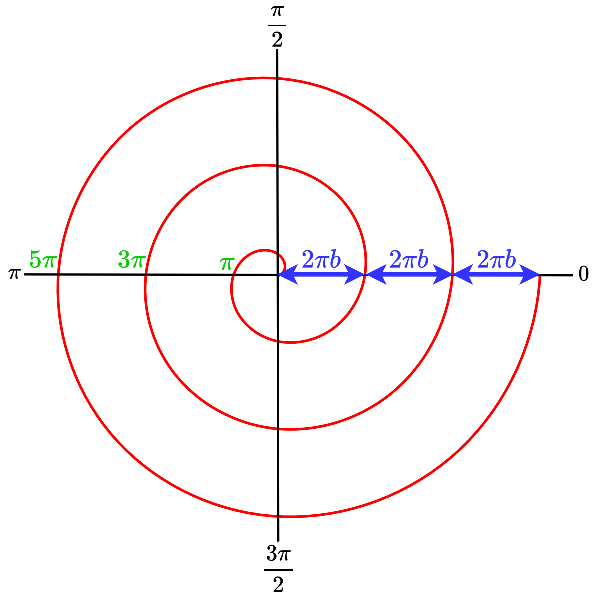

Figure 11: Diagram of Archimedean spiral used for ES and CS paths

4.4 Path parameter, velocity and altitude analysis

Starting from the intuitive parameters listed above, tests were performed at the preliminary optimal height of 640 m (obtained from POS) and preliminary search speed of 41.8 m/s (determined from initial sizing) to determine the best performing path parameters. Γ_{PS} was tested in the range [0.55, 0.75, 1.00, 1.50, 2.00], Γ_{VS} on [1.00, 1.25, 1.50, 1.75, 2.00], Γ_{CS} on [1.0, 1.2, 1.4, 1.6, 2.0] and Γ_{ES} on [0.50, 0.75, 1.00, 1.25, 1.50]. These ranges were determined based on the characteristics of each path.

A Monte Carlo simulation was run with 5, 10, 15, 20 and 24 targets (recall we can carry up to 24 buoys) to test how the performance of each parameter value varied with the number of people to save. Multiple simulations were run, with the positions of the targets randomly drawn from a uniform distribution spanning the search area until the value of J converged. Recall from Section 2 that $N_{saved}(t)$ is a random variable due to the random positions of the targets. The converged value of J was taken to be a good estimate of its expectation. For all simulations a time-step of $\Delta t = 0.1$ was used and convergence was achieved when: a minimum of 20 runs have occurred and the difference between the value of J and the rolling mean of J stayed bellow 0.1 for at least 15 consecutive simulations. The value of J obtained through the Monte Carlo simulations can thus be taken to be a good representation of the expected value. These requirements lead to a considerable number of runs before convergence, with the number varying between a couple hundred for the less demanding scenarios up to over 10'000 runs. Simulations with fewer targets were, as expected, noisier and thus required a higher number of runs to reach convergence. For reference, a single run without any plotting took around 3 seconds on an Apple M1 Pro chip with 8 CPU cores and 14 GPU cores running Matlab 2022a. Table 2 shows the results from the PS parameter analysis. Results of other analysis can be found in Appendix C.

Table 2: PS Parameter Analysis

		$\Gamma_{PS} = 0.55$	$\Gamma_{PS} = 0.75$	$\Gamma_{PS} = 1.00$	$\Gamma_{PS} = 1.50$	$\Gamma_{PS} = 2.00$
$\mathbb{E}[J]$	$\mathbb{E}[J_{N=5}]$	373	**353**	365	1747	1881
	$\mathbb{E}[J_{N=10}]$	591	574	**572**	1573	621
	$\mathbb{E}[J_{N=15}]$	802	783	**771**	1397	1480
	$\mathbb{E}[J_{N=20}]$	968	956	**936**	1202	1286
	$\mathbb{E}[J_{N=24}]$	1049	1031	**1016**	1110	1163
$\mu(\mathbb{E}[J])$		757	739	**732**	1406	1497
$\sigma(\mathbb{E}[J])$		276	278	266	**261**	289

The optimal parameters were found to be $\Gamma_{PS} = 1$, $\Gamma_{VS} = 1.25$, $\Gamma_{CS} = 1.4$ and $\Gamma_{ES} = 0.5$. The value for PS agrees with the intuition which is reassuring, however the others are slightly different. Optimal turning angle for VS ends up being close to 170°, while ES tends to perform best with significant overlap and CS with a fast convergence.

Using the path parameter values obtained from this run, all 4 paths were tested at the following altitudes $h = [200, 350, 500, 640, 750]$ m with the preliminary $v = 41.8$ m/s to find an optimal altitude. Similarly, all 4 paths were tested at 640 m with the following velocities $v = [30, 35, 41.8, 50, 58.3]$ m/s to find an optimal speed. The results can also be found in Appendix C. As expected, tests performed at smaller heights and those at lower speeds took longer to reach convergence. VS and PS ended up performing better at lower altitudes (350 m for both), while CS and ES did so at higher ones (500 m and 640 m). All paths either performed best at 58.3 m/s (cruise speed) or 41.8 m/s (preliminary search speed). The best performing CS combination was with $\Gamma_{CS} = 1.4$, $h = 500$ m and $v = 41.8$ m/s which gave a converged value of $J = 747$. ES peaked with $\Gamma_{CS} = 0.5$, $h = 640$ m and $v = 41.8$ m/s getting $J = 788$. VS performed significantly better with $J = 694$ at 350 m and 41.8 m/s and PS was by far the best with $J = 549$ at 350 m and 41.8 m/s.

While the values of J might seem high, considering that a physical interpretation of its value is the mean TTT, it is worth considering that the descent rate and ascent rate were quite small. As is further discussed in Section 5.2, for simplicity sake the UAV was taken to descend and ascend back to search altitude upon every successful detection. This is because upon detecting a target, the vehicle needs to lower its altitude down to the minimum drop height determined by Foster (which can be easily achieved via a maximum turning rate turn upon detection) in order for the drop to be accurate. However, if the vehicle remains at said altitude then the area covered by the camera will be minuscule (drop height was set at 18 m for the buoys), and so coverage of the search area would take an eternity. This constant descend and ascend procedure (coupled with a small time-window for actually dropping the payload) leads to significant increases in the TTT and in energy and fuel consumption (TTT is affected more the higher the search altitude is). Note that the values shown are for the buoys payload which required up to 24 drops. For the liferaft case, the value of J will be significantly smaller due to this mission profile only requiring a single drop.

4.5 Projected-Gradient Descent with momentum

Projected-gradient descent (PGD) with momentum was applied to the best performing path: PS with $\Gamma_{PS} = 1.0$ at 350 m altitude and 41.8 m/s. The objective function at those conditions converged to $J = 549$ in the previous section. PGD was considered due to its versatility, capability of reaching global minima (especially when paired with momentum) and ease of implementation. Compared to regular gradient-descent, PGD maps the updated weights to the set of allowable weights, therefore acting as an optimisation algorithm for constrained parameters. Recalling that altitude and flight velocity are the parameters to be analysed, weights are denoted by $w = [h, v]$. The allowable heights lied in the range $[h_{drop\ height}, h_{ceiling}]$ and the allowable velocities within $[v_{stall}, v_{cruise}]$. In most ML scenarios, analytical derivatives are used to speed up computation and decrease uncertainty, however due to the complexity of modelling the problem at hand, finite differences were used to obtain ∇J. More concretely, the 2nd order central difference approximation $\partial f'(x, y)/\partial x \approx f(x + \Delta x, y) - f(x - \Delta x, y)/(2\Delta x)$ was applied to compute approximate gradient of J. A step in the optimisation procedure involves

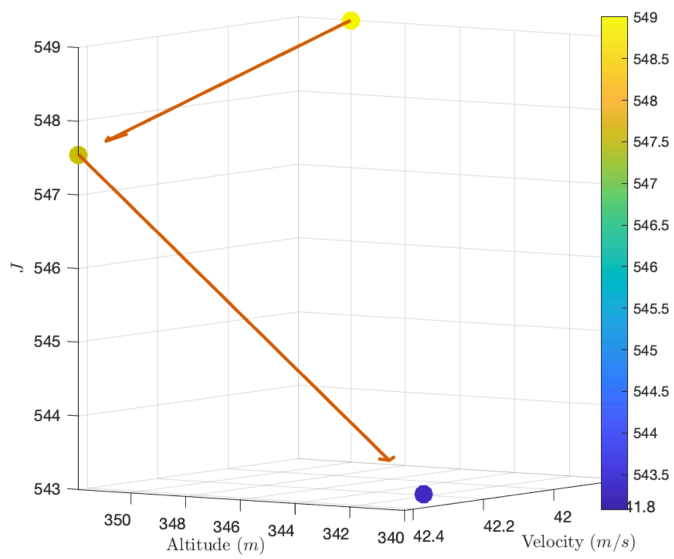

Figure 12: Gradient based optimisation of altitude and velocity for PS path

$$w_{update} = w_k - \alpha \cdot ((1 - \beta)\nabla J_k + \beta \nabla J_{k-1}) \tag{17}$$

$$w_{k+1} = \left[\min\left\{\max\{h_{drop\ height},\ w^0_{update}\},\ h_{ceiling}\right\},\ \min\left\{\max\{h_{stall},\ w^1_{update}\},\ h_{cruise}\right\}\right] \tag{18}$$

where α is the learning rate, β the momentum parameter and ∇J_k the gradient of J at step k. (18) projects the weights to satisfy their constraints. Different hyperparameter values were tested, and the final optimisation was performed with $\beta = 0.9$ and $\alpha = 0.01$. The initial guess was $w_0 = [350, 41.8]$. Figure 12 shows the best working hyperparameters which lead to $J = 543$ at 341.4 m and 42.2 m/s. Despite demonstrating a working implementation (which was further validated via testing on analytical functions) the results leave much to be desired. This has many possible explanations. Despite consistently converging to approximately close values (within 2 units of each other), equivalent runs didn't provide the exact same converged value of J. Due to the random nature of the process, and the need to obtain an estimate of the expectation, gradient values obtained via finite differences might be inaccurate. Poor performance could be explained by an highly non-convex J function, but momentum should help avoid local minima. Furthermore, since J taken to vary with h and v, 5 convergences are required per each parameter update step (4 to calculate the gradient), rendering gradient descent computationally intensive.

5 Discussion

5.1 Ethical concerns

Compared to an operator-dependent system, fully unsupervised detection should provide higher levels of reliability (faster reaction to changing or complex conditions, not susceptible to fatigue, ...). Said system might also be limited by the data with which it was trained, tested and evaluated and as such a middle-ground where the operator is only responsible for confirming detections might interesting to consider in the short term (long-term the miss-identifications done by untrained operators might become too troublesome, since the autonomous system will improve as it encounters more data). There are also ethical issues raised in the liferaft case, where a detected person might be ignored for the "greater good".

5.2 Further Work

The GC3-T camera comes equipped with a low-light visible camera which could be used instead of, or fused with, the thermal camera to aid with target detection. Visible spectrum cameras have a much greater resolution, thus allowing for flights at higher altitudes and speeds for the same detection rates. Furthermore, a fused system could improve detection by merging information from two complementary electromagnetic spectrums, and a stereo camera system allows for more accurate back-projections. Detection-wise, a tracking component could be analysed as the inclusion of temporal information has been shown to improve the identification and location accuracy [24]. Besides relying on weather forecasts to model the drift of targets, one could also use the real-time sensor measurements of the drone for that same purpose. This could be done by relating the airspeed data with the absolute position to get an estimate of wind speed, using convolutional neural networks to model sea-surface waves/currents, or more elaborate sensor fusions. While only heuristic paths were considered, further analysis could consider applying a receding horizon optimisation [43] or Reinforcement Learning [44] to allow for more intricate paths. While the UAV currently descended once per detected target, a heuristic could be introduced in the future where after each descent, the UAV would stay at the drop altitude for an tunable number of seconds, in the hopes of finding other targets nearby. Only after that number of seconds has elapsed, would it return back to search height. This would significantly decrease the time and energy costs of the search, at the cost of introducing another tunable parameter. The liferaft payload was considered when defining the objective function, through the inclusion of the discount factor, but in the future simulations could be run to test the viability of this term.

6 Conclusion

The thermal detection system described is feasible and can perform up to the requirements. The methods introduced allow for the detection of a 0.5x0.5 m wide target in a maritime setting. The camera frame-rate is more than adequate, and the resolution has been found to provide acceptable detection rates at the altitudes the UAV will be searching at. Despite the significant difference between this task and those commonly tackled by state-of-the-art CV algorithms, the problem at hand can be solved using traditional techniques.

The best search path found performs well but not up to the expected standard when deploying buoys. This was because the successive descends and ascends precluded a timely path from being possible. Gradient-based optimisations were tested but found to be too computationally intensive. Nevertheless, the Monte Carlo simulation was found to provide valuable insights into optimal path parameters, search altitudes and flight speeds, proving to be a valuable tool for MSAR scenario analysis. A definition of the MSAR problem was developed, which could be used for further path analysis down the line. Two novel paths were introduced, and despite not leading the pack performance-wise, they represent compelling alternatives if the targets were taken to be normally distributed around their LKP.

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Appendices

A Camera Fundamentals

Both in conventional and thermal cameras, the incoming radiation is converged via a bi-convex lens onto a sensor/detector (for the purposes of this report, only convex lenses are considered and so f is always positive). The focal length f (usually expressed in mm) denotes the distance between the optical centre of the lens and the sensor. f is solely a function of the lens used and is thus independent of the camera body. Let the physical dimensions of the sensor be denoted by d, for the horizontal direction, and h, for the vertical direction. The field of view (FOV), sometimes also denoted as angle of view (AOV) or angle field of view (AFOV), gives the angular view that is captured by the camera. It is both a function of the lens and the sensor, and can be calculated via:

$$FOV_H = 2 \cdot \tan^{-1}\left(\frac{d}{2f}\right) \tag{19}$$

$$FOV_V = 2 \cdot \tan^{-1}\left(\frac{h}{2f}\right) \tag{20}$$

$$FOV_D = 2 \cdot \tan^{-1}\left(\frac{\sqrt{d^2 + h^2}}{2f}\right) \tag{21}$$

where FOV_H, FOV_V, FOV_D are the horizontal, vertical and diagonal FOV respectively. Note that while FOV is usually expressed in degrees by convention, whenever it is used in a formula it must be in radians. A smaller/narrower FOV (telephoto lenses) leads to higher spatial resolution but lower field size (a wider FOV, from a wide lens, has the opposite effect). Field Size (FS) is defined as the area covered by the entire sensor (thus being a function of FOV, distance to target and sensor resolution). Let us introduce ground sample distance (GSD) which indicates the area seen by a single pixel of the image sensor. Let the camera resolution be denoted by $(H_{\text{px}}, V_{\text{px}})$, where H_{px} is an integer denoting the number of horizontal pixels and V_{px} one denoting the number of vertical pixels in the sensor. Then the horizontal and vertical GSD are given respectively by

$$GSD_H = 2 \cdot \tan\left(\frac{FOV_H}{2 \cdot H_{\text{px}}}\right) \cdot R \tag{22}$$

$$GSD_V = 2 \cdot \tan\left(\frac{FOV_V}{2 \cdot V_{\text{px}}}\right) \cdot \frac{R}{\cos(\theta)} \tag{23}$$

$$\tag{24}$$

where θ is the camera's tilt angle, such that when $\theta = 0°$ the camera is pointing directly downwards and when $\theta = 90°$ the camera is pointing directly forwards (along the aircraft's datum). R denotes the distance between the camera and the target. Figure A.1 illustrates these variables. The greater the number of pixels in the sensor is, the better the spatial resolution.

Using GSD, we can calculate FS by

$$FS_H = GSD_H \cdot H_{\text{px}} \tag{25}$$

$$FS_V = GSD_V \cdot V_{\text{px}} \tag{26}$$

$$\tag{27}$$

To sum up, the higher the search altitude, the lower the spatial resolution will be (a single pixel corresponds to a greater ground area) but the larger FS will be (a larger ground area is covered by the camera). Note that, despite often being misused, FOV solely indicates the angular extent of the scene captured by the image.

For a given characteristic dimension d_c and average ground sample distance GSD_{avg}, the number of cycles across the target N is given by

$$N = \frac{d_c}{2 \cdot GSD_{avg}} \tag{28}$$

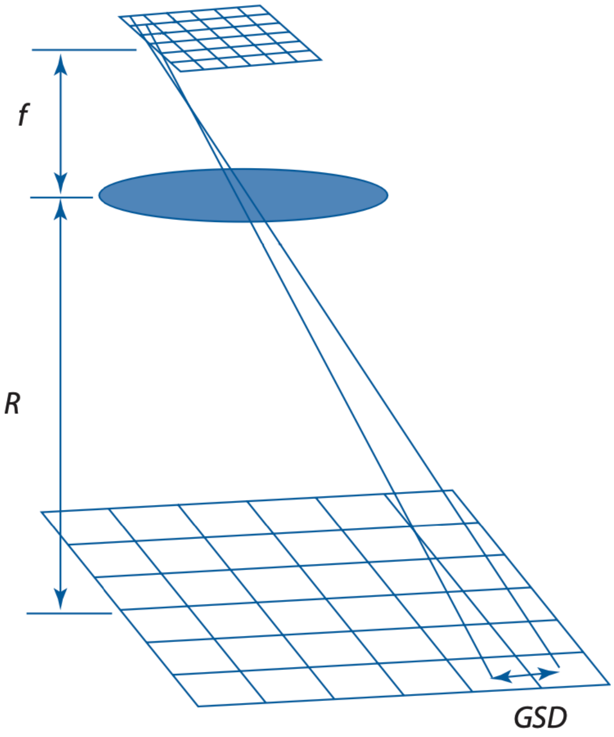

Figure A.1: GSD Geometry, modified from [15]

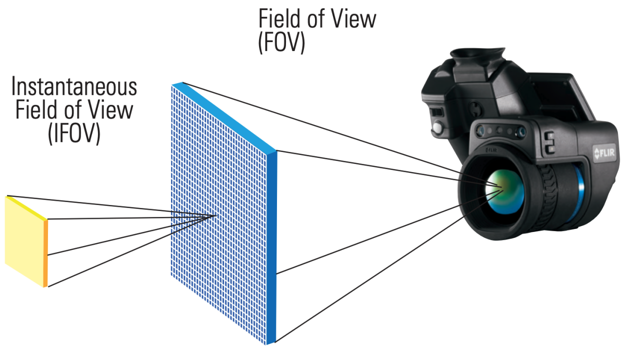

Figure A.2: IFOV, from [11]

B POC and POD

Below is POC and POD as a function of the UAV's altitude.

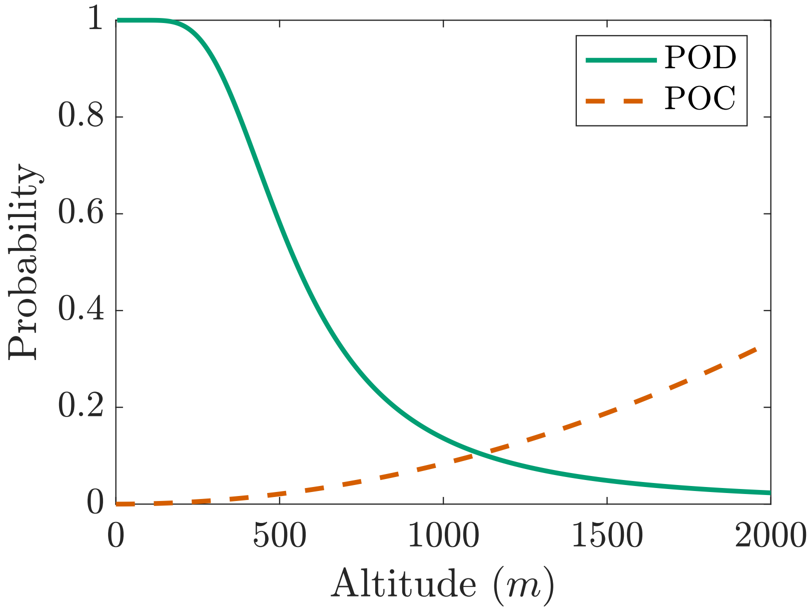

Figure B.1: POC and POD with altitude

C Path velocity and altitude analysis

Table C.1: PS Altitude Analysis

		$h = 200$ m[1]	$h = 350$ m	$h = 500$ m	$h = 640$ m	$h = 750$ m
$\mathbb{E}[J]$	$\mathbb{E}[J_{N=5}]$	-	**307**	313	372	372
	$\mathbb{E}[J_{N=10}]$	-	**425**	481	579	623
	$\mathbb{E}[J_{N=15}]$	-	**549**	650	769	857
	$\mathbb{E}[J_{N=20}]$	-	**667**	818	933	1003
	$\mathbb{E}[J_{N=24}]$	-	**795**	925	1012	1076
$\mu(\mathbb{E}[J])$		-	**549**	638	733	786
$\sigma(\mathbb{E}[J])$		-	**192**	247	261	289

Table C.2: PS Velocity Analysis

		$v = 30$ m/s	$v = 35$ m/s	$v = 41.8$ m/s	$v = 50$ m/s	$v = 58.3$ m/s
$\mathbb{E}[J]$	$\mathbb{E}[J_{N=5}]$	**367**	392	377	401	406
	$\mathbb{E}[J_{N=10}]$	581	578	**572**	583	579
	$\mathbb{E}[J_{N=15}]$	794	782	766	766	**757**
	$\mathbb{E}[J_{N=20}]$	961	947	929	922	**916**
	$\mathbb{E}[J_{N=24}]$	1038	1028	1014	1007	**997**
$\mu(\mathbb{E}[J])$		748	745	732	736	**731**
$\sigma(\mathbb{E}[J])$		275	261	260	247	**241**

[1] At 200 m altitude and 41.8 m/s flight speed, the turning radius required to keep $\Gamma_{PS} = 1.00$ is smaller than the UAV's minimum turning radius. As such performance at this altitude could not be evaluated.

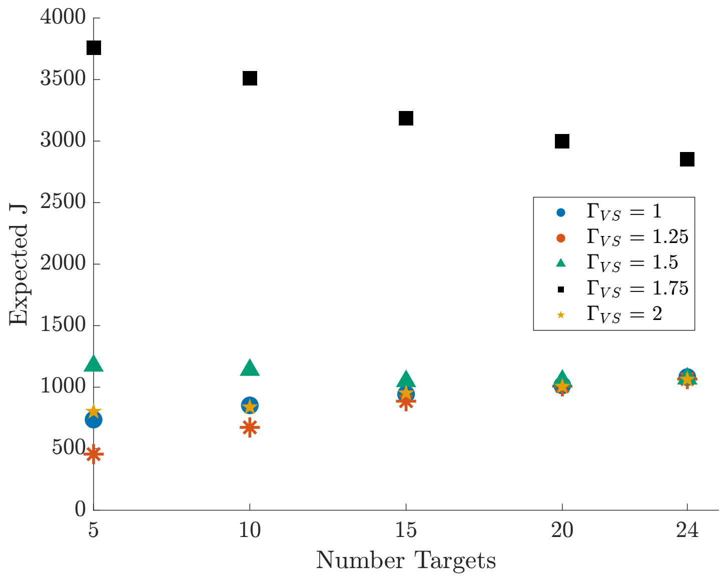

Figure C.1: VS Parameter Analysis

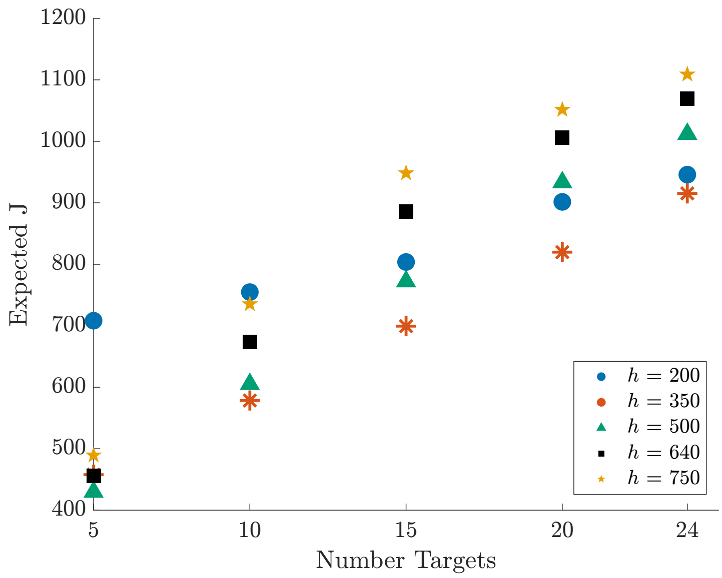

Figure C.2: VS Altitude Analysis

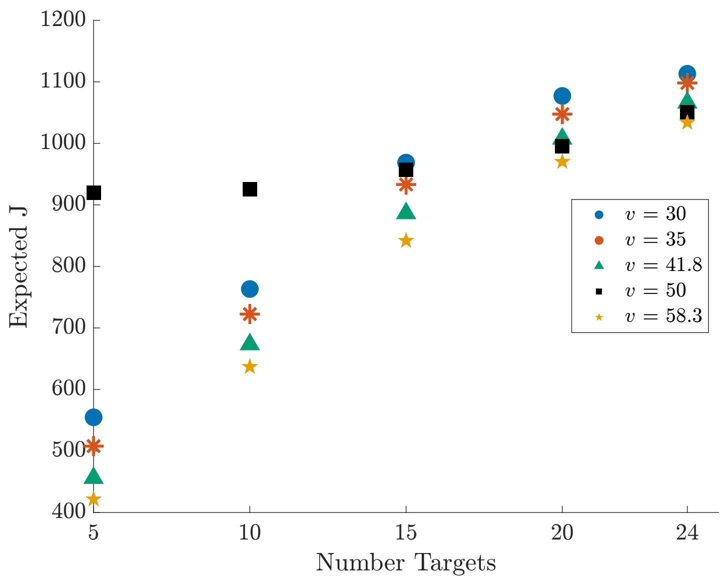

Figure C.3: VS Velocity Analysis

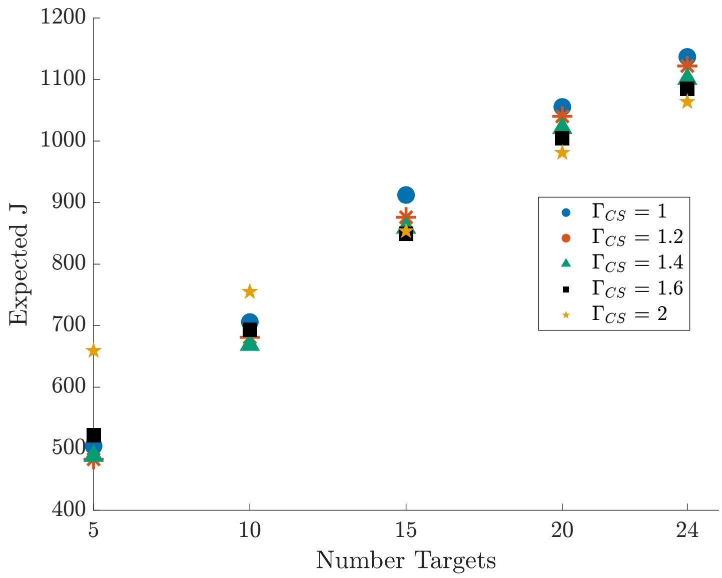

Figure C.4: CS Parameter Analysis

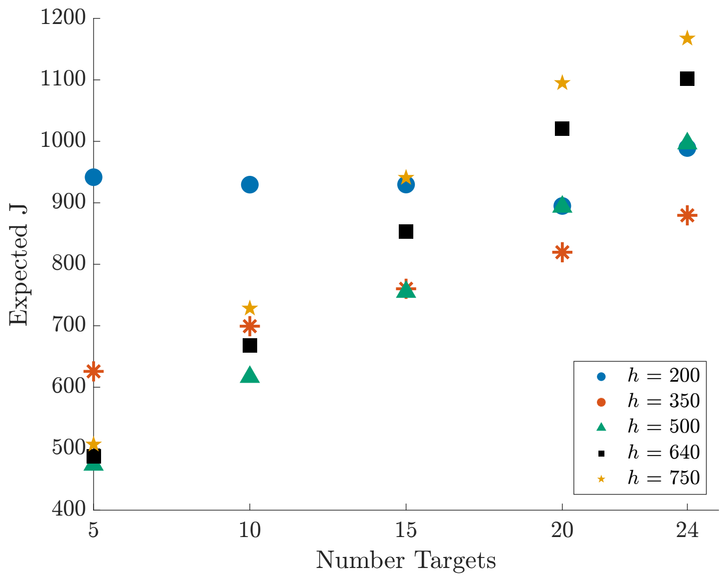

Figure C.5: CS Altitude Analysis

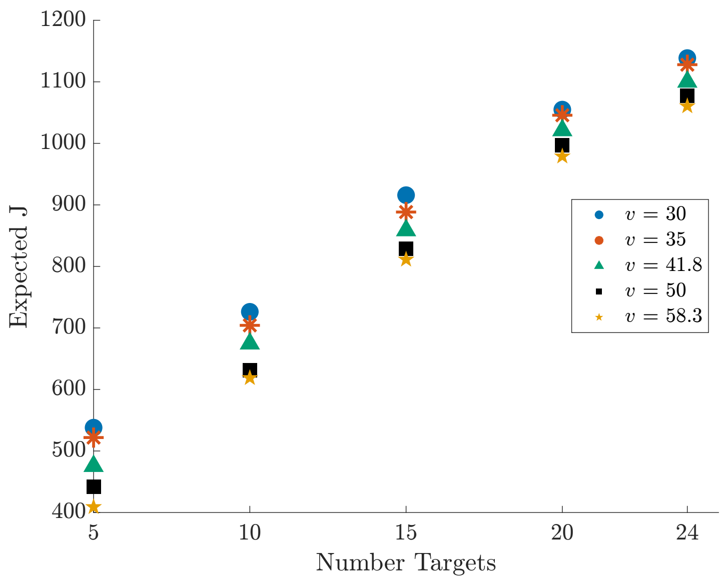

Figure C.6: CS Velocity Analysis

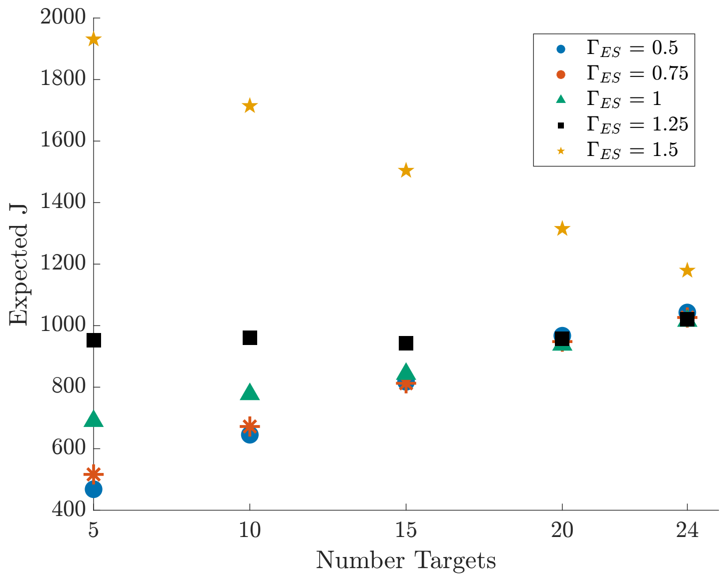

Figure C.7: ES Parameter Analysis

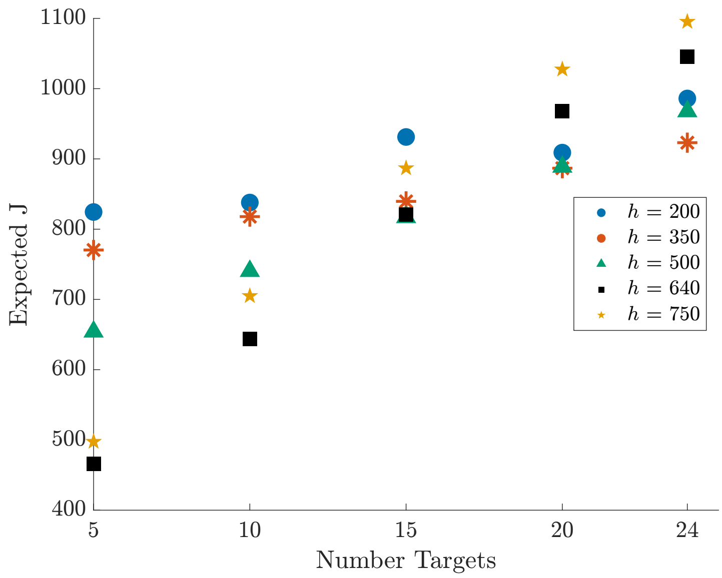

Figure C.8: ES Altitude Analysis

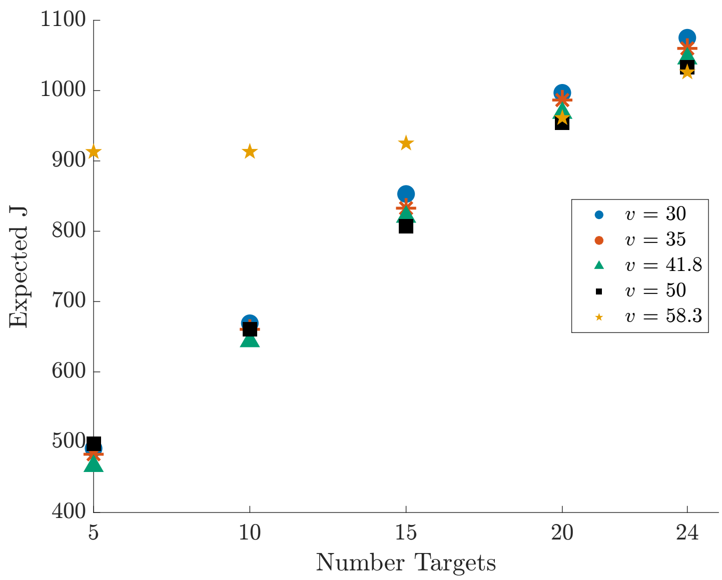

Figure C.9: ES Velocity Analysis